ITSE 2321
Object-Oriented Programming
Exam 1 - Part 2:
Computer Program #1

THIS PART OF THE EXAM IS WORTH - <u>35 Points</u>

<u>The Movie Ticket and Snack Sales Program:</u>

Create a C# computer program that will compute the movie ticket and snack sale for a customer.

****Detail instructions start on page 3.**

Your form, the user interface, will have a listbox containing 4 different ticket prices, which are:

Senior (55 years old and up - $5.50)
Child (Under the age of 5 - $6.25)
Matinee ($6.50, before 6 P.M.)
Adult ($8.75)

In addition, your form will have 3 labels:

Number of Tickets:
Total Cost:
And a label that contains the total cost of the movie ticket and snack sale for the customer in currency.

3 checkboxes for:

Soda ($4.25)
Popcorn ($3.00)
Candy ($2.50)

Also, your form will have a text box, which is use to enter the number of tickets for the customer.

And one PictureBox control that contains the "Popcorn" image file.

The calculate button will determine the total cost of the movie tickets along with the snack sale, which will be displayed in your total cost label in currency.

In addition, display a message in a message box if the user forgets to input the number of tickets, inputs text, like "dog", for the textbox OR if the user does <u>not</u> select a ticket price from the listbox.

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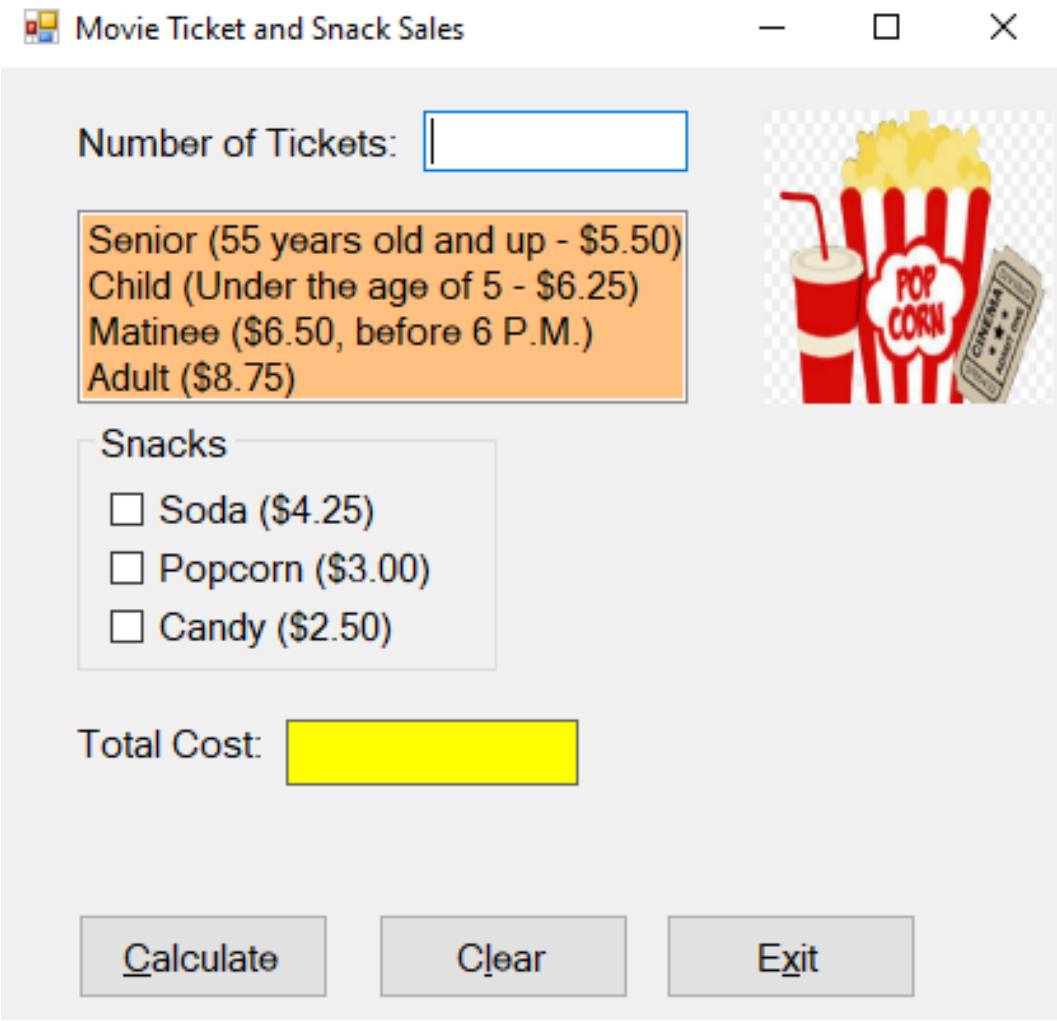
The clear button will clear out the textbox, the user's selection in the listbox, the three checkboxes for the snacks, and the total cost label. In addition, the clear button will reset any variables in your computer program to 0. Also, the clear button will reset the focus to the textbox for the next customer.

*The naming of your objects and variables should reflect the purpose of the controls on your interface and the variables in your computer code. Don't forget your comments (programmer's remarks) and error-trapping. Be consistent, stick with one standard.

Sample Interface:

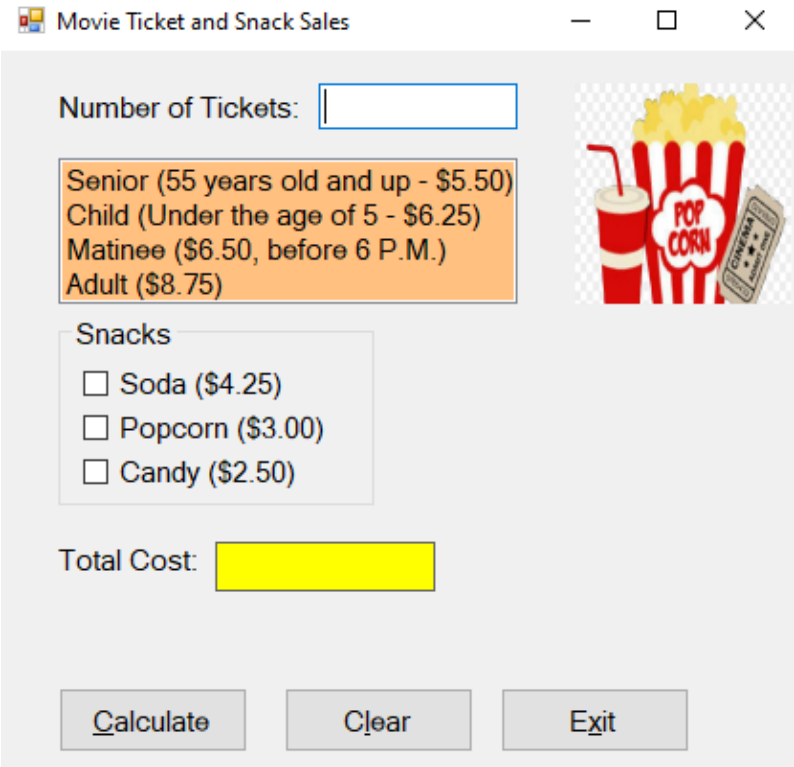


*Name your project with your name and the word "Exam1" - for example – PatrickLogueExam1. Save your project often. When you are finish with your computer program; make sure you **save** the final version of it. Compress the computer program's folder and take the compressed file and upload it into Blackboard's "Quizzes and Exams" Section, Exam 1 - Part 2: Computer Program 1.

Part A: 10 Points: Create the User Interface – The Form

Create the user interface from scratch.

***Hint: You can copy the text from this document and paste it into your control's properties. This will save you some time in creating the user interface.**



1. Create the form described and pictured above (3 labels, 1 textbox, 1 listbox, 1 Groupbox with the text "Snacks" that contains 3 check boxes for the 3 different types of snacks along with their prices, 1 picturebox that contains the provided "Popcorn" image (SizeMode = StretchImage), and 3 buttons: Calculate, Clear, and Exit). Again, use appropriate labels to identify items on the interface.
2. Make the textbox's BackColor white and the total cost label's BackColor yellow and the ForeColor black (not the labels, like "Number of Tickets", that are used to identify the items on the interface).
3. Name your form and give it a caption – like "Movie Ticket and Snack Sales"

Part B: 5 Points: Exit and Clear Buttons

1. Code the exit button to close the program. Make the x in Exit an access key for the button (see page 179).
2. Code the clear button to clear out the textbox, the user's selection in the listbox, the three checkboxes for the snacks, and the total cost label. In addition, your clear button will reset any variables in your computer program to 0. Also, the clear button will put the insert point back into the textbox (give it focus) for the next customer. Make the l in Clear an access key for the button.

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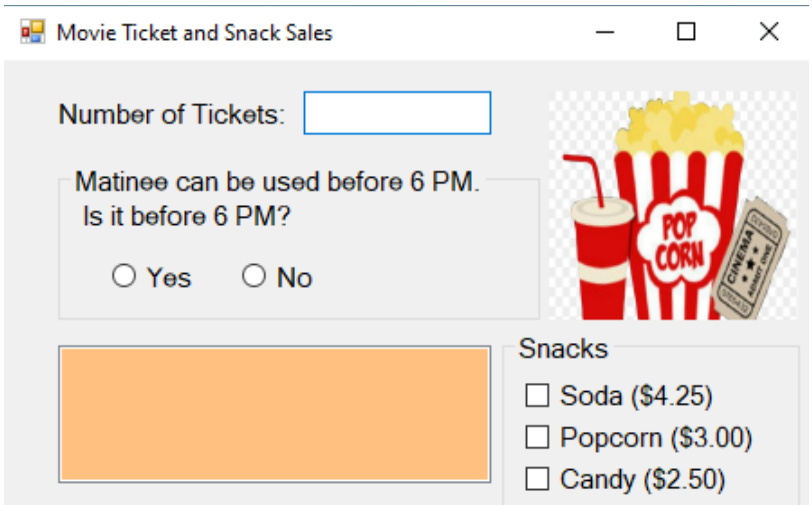
Part C: 20 Points: Calculations

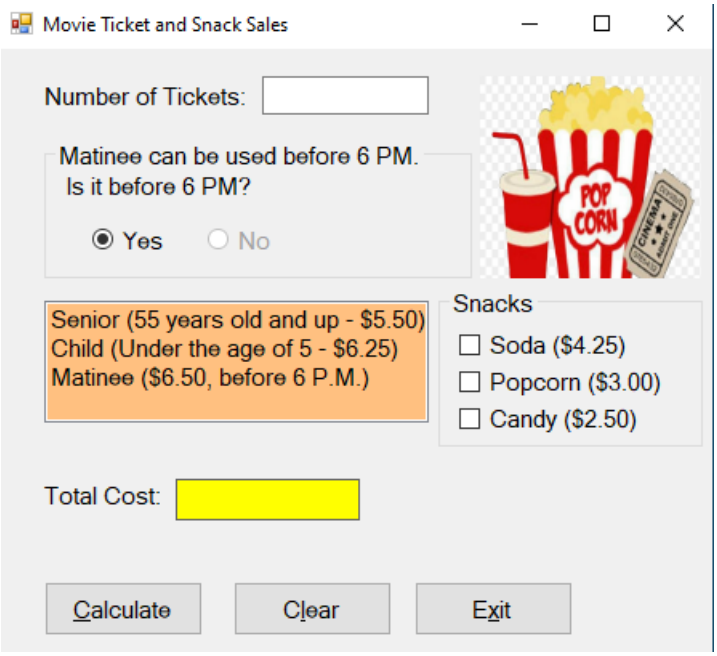
1. Code the calculate button. Make the C in Calculate an access key for the button.
2. Add appropriate variables to your computer program. Place these variables or define them in the appropriate places (most efficient and effective way – for the program and for the computer's system resources and memory). The information that is collected from the user's input during run time should be used to calculate the movie ticket and snack sale transaction.
3. Make sure that the user inputs a number in the textbox. Make sure that the data is numeric for the textbox. Do not let the user type in words, like "dog", in the textbox – Catch this error and send a message to the user to indicate the error. Clear and reset the focus in the textbox (Chapter 4's Validating the Data in a Textbox = Input Validation).
4. In addition, display a message in a message box if the user does not select a ticket price from the listbox.
5. Determine which ticket price was selected from the listbox and calculate the customer's movie ticket sale (price times the number of tickets).
6. You will also need to write the code that will determine which checkboxes were selected for the snacks and process the appropriate snack transaction (Soda, Popcorn, Candy, All of them or individually). Calculate the customer's snack sale.
7. Add together the customer's movie ticket sale and snack sale and display the total cost in a label. Format the total cost in the label as currency.

****Extra Credit: 5 Points:** Determine if customer can purchase a matinee ticket

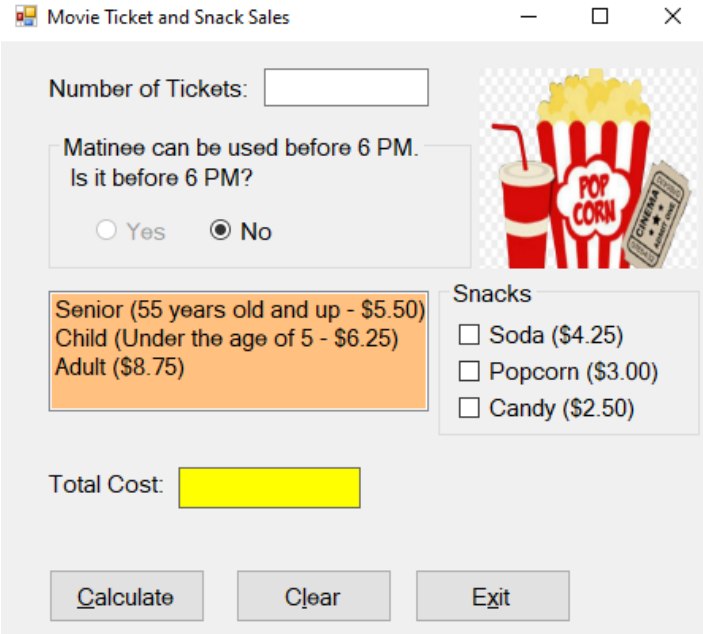


1. Modify the user interface by adding a Groupbox with the text "Matinee can be used before 6 PM.", which contains 1 label "Is it before 6 PM?" and 2 radio buttons: Yes and No.

2. If the user states “Yes” (selecting the “Yes” radio button), it is before 6 PM; then display only the senior, child, and matinee ticket prices in the listbox. Also, disable the “No” radio button. See sample interface below:



3. If the user states “No” (selecting the “No” radio button), it is after 6 PM; then display only the senior, child, and adult ticket prices in the listbox. Also, disable the “Yes” radio button. See sample interface below:



4. You would also need to add some code in the Clear button, which will enable (make active) the two radio buttons: Yes and No for the next customer and set the checked property of these 2 radio buttons to false. Also, you may need to clear out the listbox or not.

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